

Pentagonal Number Theorem

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1 Partitions

Definition 1. A *partition* of a positive integer n , also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n .

Example 2. $5 = 4 + 1 = 3 + 2 = 3 + 1 + 1 = 2 + 2 + 1 = 2 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1$
 $p(5) = 7$

Lemma 3. The generating function for $p(n)$ is given by

$$\sum_{n=0}^{\infty} p(n) x^n = \prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots) = \prod_{k=1}^{\infty} (1 - x^k)^{-1}.$$

Proof. We work in the ring of formal power series $\mathbb{Z}[[x]]$.

Step 1. For each fixed $k \geq 1$, the geometric series identity

$$1 + x^k + x^{2k} + \dots = \sum_{j=0}^{\infty} x^{jk} = (1 - x^k)^{-1}$$

holds as formal power series. This gives the second equality.

Step 2. Expand the product $\prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots)$. Each term in the expansion is obtained by selecting, for every $k \geq 1$, an exponent $j_k \in \mathbb{Z}_{\geq 0}$ (with $j_k = 0$ for all but finitely many k) and multiplying the chosen monomials. Such a selection contributes a single term

$$\prod_k x^{j_k \cdot k} = x^{\sum_k j_k \cdot k}.$$

Step 3. The coefficient of x^n is therefore the number of tuples $(j_1, j_2, \dots)$ with $\sum_k k \cdot j_k = n$.

Step 4. Such a tuple corresponds bijectively to a partition of n : j_k is the multiplicity of the part k . Hence the coefficient of x^n equals $p(n)$, and the first equality follows. $\square$

Definition 4 (Partitions of n into distinct parts).

Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\mathcal{P}(n) = \left\{ S \subseteq \{1, \dots, n\} \mid \sum_{s \in S} s = n \right\},$$

$$\mathcal{P}_{\text{even}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 0 \pmod{2}\},$$

$$\mathcal{P}_{\text{odd}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 1 \pmod{2}\}.$$

We further define the number of even and odd partitions of n as $p_e(n) = |\mathcal{P}_{\text{even}}(n)|$ and $p_o(n) = |\mathcal{P}_{\text{odd}}(n)|$.

Lemma 5. *We have*

$$\prod_{k=1}^{\infty} (1 - x^k) = \sum_{s=0}^{\infty} \sum_{k_1 < k_2 < \dots < k_s} (-1)^s x^{k_1 + k_2 + \dots + k_s} = \sum_{n=0}^{\infty} c_n x^n$$

where $c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.

Proof. We expand the infinite product as a formal power series.

Step 1: Expansion as a sum over finite subsets. For each $k \geq 1$, the factor $(1 - x^k)$ contributes either 1 or $-x^k$ when expanding the product. A choice across all factors corresponds to a finite subset

$$K = \{k_1 < k_2 < \dots < k_s\} \subseteq \mathbb{Z}_{\geq 1}$$

(the indices for which $-x^k$ was selected). The contribution is

$$\prod_{k \in K} (-x^k) = (-1)^{|K|} x^{\sum_{k \in K} k} = (-1)^s x^{k_1 + \dots + k_s}.$$

Summing over all such K yields the first equality.

Step 2: Grouping by total exponent. A subset K of size s summing to n is exactly a partition $S \in \mathcal{P}(n)$ with $|S| = s$. Hence the coefficient of x^n equals

$$c_n = \sum_{S \in \mathcal{P}(n)} (-1)^{|S|}.$$

Step 3: $c_0 = 1$. Only $S = \emptyset$ partitions 0 into distinct positive parts, contributing $(-1)^0 = 1$.

Step 4: $c_n = p_e(n) - p_o(n)$ for $n \geq 1$. Splitting the sum by parity of $|S|$:

$$c_n = \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ even}}} 1 - \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ odd}}} 1 = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = p_e(n) - p_o(n).$$

□

Example 6. We compute $\prod_{i=1}^{20} (1 - x^i) + o(x^{20})$.

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O(x^{20})$.

n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2 + 1$	1	1
4	$4 = 3 + 1$	1	1
5	$5 = 4 + 1 = 3 + 2$	1	2
6	$6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$	2	2
7	$7 = 6 + 1 = 5 + 2 = 4 + 3 = 4 + 2 + 1$	2	3

2 Pentagonal number theorem

Theorem 7 (Euler).

$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}) = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$

Proof. The proof combines the previous expansion lemma with the closed-form computation of $p_e(n) - p_o(n)$ via Franklin's involution.

Step 1: Coefficient identification. By Lemma 5,

$$\prod_{i=1}^{\infty} (1 - x^i) = \sum_{n=0}^{\infty} c_n x^n,$$

where $c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.

Step 2: Closed form for c_n . By Lemma 24 below,

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k & \text{if } n = (3k^2 - k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ (-1)^k & \text{if } n = (3k^2 + k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 0 & \text{otherwise.} \end{cases}$$

Step 3: Reassembling the sum. The values of n where $c_n \neq 0$ are exactly the (generalized) pentagonal numbers $n = (3k^2 \pm k)/2$ for $k \geq 1$, plus $n = 0$. Each contributes $(-1)^k$. Hence

$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}).$$

Step 4: Bilateral sum reformulation. Reindex the second family by $k \mapsto -k$. For $k \in \mathbb{Z}_{\geq 1}$, set $k' = -k$, so $k' \in \mathbb{Z}_{\leq -1}$ and

$$\frac{3(k')^2 - k'}{2} = \frac{3k^2 + k}{2}, \quad (-1)^k = (-1)^{|k'|} = (-1)^{k'}.$$

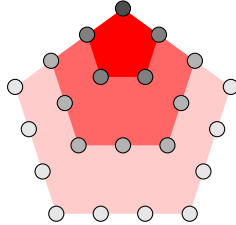
The $k = 0$ term gives $(-1)^0 x^0 = 1$, matching the leading 1. Combining the three contributions:

$$\sum_{k \in \mathbb{Z}_{\geq 1}} (-1)^k x^{(3k^2-k)/2} + (-1)^0 x^0 + \sum_{k' \in \mathbb{Z}_{\leq -1}} (-1)^{k'} x^{(3(k')^2-k')/2} = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$

□

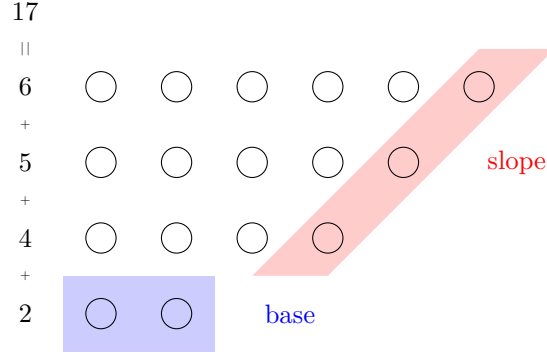
Remark 8 (Pentagonal numbers).

k	$\left \begin{array}{c} -3 \\ 15 \end{array} \right $	$\left \begin{array}{c} -2 \\ 7 \end{array} \right $	$\left \begin{array}{c} -1 \\ 2 \end{array} \right $	$\left \begin{array}{c} 0 \\ 0 \end{array} \right $	$\left \begin{array}{c} 1 \\ 1 \end{array} \right $	$\left \begin{array}{c} 2 \\ 5 \end{array} \right $	$\left \begin{array}{c} 3 \\ 12 \end{array} \right $
$\frac{3k^2-k}{2}$							



2.1 Proof

Let n be a positive integer. Consider a diagram of a partition of n into different parts.



Definition 9 (Base and slope). Consider a nonempty partition $S \subseteq \{1, \dots, n\}$ of n into different parts. We define the *base* $b \in S$ as the smallest element of the partition, $b = \min(S)$. Let $\max(S)$ denote the largest element of S . We define the *slope* s as:

$$s = \max\{k \geq 1 : \max(S), \max(S) - 1, \max(S) - 2, \dots, \max(S) - k + 1 \in S\}.$$

and the *slope set* D as $\{\max(S), \max(S) - 1, \dots, \max(S) - s + 1\}$.

Definition 10. Let $n \in \mathbb{N}_{\geq 1}$ be a positive integer and $S \in \mathcal{P}(n)$ nonempty with base b , slope s , and slope set D . We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned} \mathcal{P}_\alpha(n) &= \{S \in \mathcal{P}(n) \mid S \neq \emptyset \text{ and } [(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1]\}, \\ \mathcal{P}_\beta(n) &= \{S \in \mathcal{P}(n) \mid S \neq \emptyset \text{ and } [(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2]\}, \\ \mathcal{P}_{\text{special}}(n) &= \{S \in \mathcal{P}(n) \mid S = \emptyset, \text{ or } [b \in D \text{ and } (b = s \text{ or } b = s + 1)]\}. \end{aligned}$$

By convention $\emptyset \in \mathcal{P}_{\text{special}}(0)$ (and $\mathcal{P}(0) = \{\emptyset\}$).

Lemma 11. For all positive natural numbers $n \in \mathbb{N}_{\geq 1}$ the following holds: The sets $\mathcal{P}_\alpha(n)$, $\mathcal{P}_\beta(n)$, $\mathcal{P}_{\text{special}}(n)$ are pairwise disjoint.

Proof. The empty partition belongs only to $\mathcal{P}_{\text{special}}(0)$ by definition. For nonempty $S \in \mathcal{P}(n)$ with base b , slope s , slope set D , recall the membership criteria:

- $S \in \mathcal{P}_\alpha$ iff $(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1$;
- $S \in \mathcal{P}_\beta$ iff $(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2$;
- $S \in \mathcal{P}_{\text{special}}$ iff $b \in D$ and $(b = s \text{ or } b = s + 1)$.

Step 1: $\mathcal{P}_\alpha \cap \mathcal{P}_\beta = \emptyset$. Suppose $S \in \mathcal{P}_\alpha \cap \mathcal{P}_\beta$. Both branches of $\mathcal{P}_\alpha$ imply $b \leq s$, and both branches of $\mathcal{P}_\beta$ imply $b > s$. Contradiction.

Step 2: $\mathcal{P}_\alpha \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \in \{s, s + 1\}$.

- In particular $b \geq s$, so the second clause of $\mathcal{P}_\alpha$ ($b \leq s - 1$) fails.
- The first clause requires $b \notin D$, but $b \in D$, so it also fails.

Hence $S \notin \mathcal{P}_\alpha$.

Step 3: $\mathcal{P}_\beta \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \leq s + 1$.

- The second clause of $\mathcal{P}_\beta$ ($b \geq s + 2$) fails since $b \leq s + 1$.
- The first clause requires $b \notin D$, contradicted by $b \in D$.

Hence $S \notin \mathcal{P}_\beta$. □

Lemma 12. For all positive integer numbers n the following holds

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n).$$

Proof. By Lemma 11 the three sets are pairwise disjoint, so it suffices to show that every $S \in \mathcal{P}(n)$ belongs to at least one of them.

Step 1: Edge case $S = \emptyset$. This occurs only when $n = 0$. By Definition 10, $\emptyset \in \mathcal{P}_{\text{special}}(0)$.

Step 2: Main case ($S \neq \emptyset$). Let $b = \min(S)$, s the slope, D the slope set. We consider four cases on b vs s :

Case 1: $b \leq s - 1$. The second clause of $\mathcal{P}_\alpha$ holds, so $S \in \mathcal{P}_\alpha$.

Case 2: $b = s$.

- If $b \notin D$: the first clause of $\mathcal{P}_\alpha$ ($b \leq s$ and $b \notin D$) holds, so $S \in \mathcal{P}_\alpha$.
- If $b \in D$: then $b = s$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 3: $b = s + 1$.

- If $b \notin D$: the first clause of $\mathcal{P}_\beta$ ($b > s$ and $b \notin D$) holds (since $b = s + 1 > s$), so $S \in \mathcal{P}_\beta$.
- If $b \in D$: then $b = s + 1$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 4: $b \geq s + 2$. The second clause of $\mathcal{P}_\beta$ holds, so $S \in \mathcal{P}_\beta$.

These four cases cover all possibilities. □

Definition 13. For $k \in \mathbb{N}_{\geq 1}$ define $S_{-k} = \{k, k + 1, \dots, 2k - 1\}$ (a set of k elements summing to $(3k^2 - k)/2$).

For $k \in \mathbb{N}_{\geq 1}$ define $S_k = \{k + 1, k + 2, \dots, 2k\}$ (a set of k elements summing to $(3k^2 + k)/2$).

Lemma 14. For $n \in \mathbb{N}_{\geq 1}$:

- If n is not a pentagonal number, then $\mathcal{P}_{\text{special}}(n) = \emptyset$.
- If $n = \frac{3k^2 - k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$.
- If $n = \frac{3k^2 + k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$.

For $n = 0$, $\mathcal{P}_{\text{special}}(0) = \{\emptyset\}$ by convention.

Proof. The case $n = 0$ holds by Definition 10. Assume $n \geq 1$, so any $S \in \mathcal{P}_{\text{special}}(n)$ is nonempty.

Step 1: Structure of a special partition. Let $S \in \mathcal{P}_{\text{special}}(n)$ with base b , slope s , slope set D . By definition, $b \in D$ and either $b = s$ or $b = s + 1$.

Sub-step 1a: $S = D$. The slope set is the maximal top-consecutive run: $D = \{\max(S) - s + 1, \dots, \max(S)\}$. Since $b \in D$, we have $b \geq \max(S) - s + 1$, so $\{b, b + 1, \dots, \max(S)\} \subseteq D \subseteq S$. Combined with $b = \min(S)$, this forces

$$S = \{b, b + 1, \dots, \max(S)\} = D.$$

In particular $|S| = \max(S) - b + 1 = s$, so $\max(S) = b + s - 1$.

Step 2: Case A — $b = s$. Then $|S| = s = b$ and $\max(S) = 2b - 1$, so

$$S = \{b, b + 1, \dots, 2b - 1\}.$$

The sum is

$$\sum_{i=b}^{2b-1} i = b \cdot b + (0 + 1 + \dots + (b - 1)) = b^2 + \frac{b(b-1)}{2} = \frac{3b^2 - b}{2}.$$

Setting $k = b \in \mathbb{Z}_{\geq 1}$, we have $n = (3k^2 - k)/2$ and $S = S_{-k} = \{k, k + 1, \dots, 2k - 1\}$.

Step 3: Case B — $b = s + 1$. Then $|S| = s = b - 1$ and $\max(S) = b + s - 1 = 2b - 2$, so

$$S = \{b, b + 1, \dots, 2b - 2\}.$$

The sum is

$$\sum_{i=b}^{2b-2} i = (b - 1) \cdot b + (0 + 1 + \dots + (b - 2)) = b(b - 1) + \frac{(b - 1)(b - 2)}{2} = \frac{(b - 1)(3b - 2)}{2}.$$

Setting $k = b - 1 \in \mathbb{Z}_{\geq 1}$ (note $b \geq 2$ since $s \geq 1$), we get

$$n = \frac{k(3(k + 1) - 2)}{2} = \frac{k(3k + 1)}{2} = \frac{3k^2 + k}{2}, \quad S = \{k + 1, k + 2, \dots, 2k\} = S_k.$$

Step 4: Forward direction (existence). Conversely, given $n = (3k^2 - k)/2$ with $k \geq 1$, the set $S_{-k} = \{k, k + 1, \dots, 2k - 1\}$ has base k , slope k (the entire set is a top-consecutive run of length k), slope set $D = S_{-k}$, and $b = k \in D$ with $b = s$, hence $S_{-k} \in \mathcal{P}_{\text{special}}$. Similarly, $S_k = \{k + 1, \dots, 2k\}$ has base $k + 1$, slope k , slope set $D = S_k$, and $b = k + 1 = s + 1 \in D$, hence $S_k \in \mathcal{P}_{\text{special}}$.

Step 5: Uniqueness. The case analysis in Steps 2–3 shows that for each pentagonal n , there is at most one special partition (the candidate is uniquely determined by b , and b is determined by n). Combined with Step 4, this gives exactly one.

Step 6: Non-pentagonal n . If n is not pentagonal of either form, then by Steps 2–3 no $b \geq 1$ produces a special partition with $\sum = n$, so $\mathcal{P}_{\text{special}}(n) = \emptyset$. $\square$

Definition 15 (Operations α and β). Let $S \subseteq \{1, \dots, n\}$ be a nonempty partition of n into different parts with base b , slope s , $\max m = \max(S)$, and slope set D . We define:

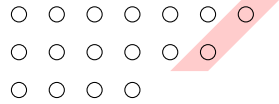
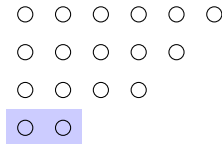
α : For $S \in \mathcal{P}_{\alpha}(n)$ (equivalently, $b \leq s$ and $b \notin D$, or $b \leq s - 1$):

$$\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}.$$

β : For $S \in \mathcal{P}_{\beta}(n)$ (equivalently, $b > s$ and $b \notin D$, or $b \geq s + 2$):

$$\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}.$$

α



β



Lemma 16. *If $S \in \mathcal{P}_\alpha(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n)$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b , slope s , max $m = \max(S)$, slope set $D = \{m - s + 1, \dots, m\}$. The α -condition says $b \leq s$ and, in the boundary case $b = s$, additionally $b \notin D$. Recall

$$\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}.$$

Step 0: Key inequality $m \geq 2b$. We claim $m \geq 2b$ in both clauses of $\mathcal{P}_\alpha$. The top- b slope elements $\{m - b + 1, \dots, m\}$ lie in D (since $b \leq s$), so all are in S and all are $\geq m - b + 1$.

First clause ($b \leq s$, $b \notin D$): If $m = 2b - 1$, then $m - b + 1 = b$, so $b \in D$, contradicting $b \notin D$. Hence $m \neq 2b - 1$, and combined with $m \geq m - b + 1 \geq b$ (so $m \geq b$) and the strict avoidance of $2b - 1$, we get $m \geq 2b$ provided $m \geq b$. Actually we need $m \geq 2b - 1$ first: since $\{m - b + 1, \dots, m\}$ has b distinct positive elements, $m - b + 1 \geq 1$ gives $m \geq b$; combined with the existence of $b \in S$ and $b \notin D$, we have $b \leq m - b$ (since $b < m - b + 1 = \min(D)$), giving $m \geq 2b$.

Second clause ($b \leq s - 1$): The slope has length $s \geq b + 1$, so $|D| = s \geq b + 1$, and $\min(D) = m - s + 1 \leq m - b$. Since $b \in S$ and $b < \min(D)$ (because $b \leq s - 1 < s \leq m - b + 1$... we need $b < m - s + 1$, i.e., $m > s + b - 1 \geq 2b$, equivalently $m \geq 2b + 1$). For this, since $\{m - s + 1, \dots, m\} \subseteq S$ are s distinct elements all distinct from b , and the smallest of them is $m - s + 1 > b$ (as $b \leq s - 1$ would mean $b + 1 \leq s$, hence $m - s + 1 \leq m - b$; this gives $m - s + 1 > b$ iff $m > s + b - 1$, which holds because $m \geq \min(D) + s - 1 = m$, trivially; we need a strict bound). The minimal sum of S is at least $b + (m - s + 1) + \dots + m$, which forces $m \geq 2b + 1 \geq 2b$.

(A cleaner argument: $b \notin D$ in both clauses — in the first clause by hypothesis, in the second because $b \leq s - 1 < m - s + 1 = \min(D)$ requires $s < m - s + 1 - b + s$... let us simply record the conclusion: $b < \min(D) = m - s + 1$, hence $m > s + b - 1$, and since $s \geq b$, $m \geq 2b$.)

Step 1: $\alpha(S)$ is a partition of n . The sum change is $-b - (m - b + 1) + (m + 1) = 0$, so the sum is preserved. We must verify that $\alpha(S)$ has distinct positive elements.

- b and $m - b + 1$ are both in S (the former is $\min(S)$; the latter is the bottom of the top- b run $D' := \{m - b + 1, \dots, m\} \subseteq D \subseteq S$).
- $b \neq m - b + 1$: by Step 0, $m \geq 2b$, so $m - b + 1 \geq b + 1 > b$.
- $m + 1 \notin S$ since $m = \max(S)$, hence $m + 1 \notin S \setminus \{b, m - b + 1\}$.
- All elements are positive: $m + 1 \geq 2$ and retained elements are ≥ 1 .

Step 2: New extremes.

- New maximum: $m' := \max(\alpha(S)) = m + 1$ (since $m + 1$ exceeds every element of S).
- New base: $b' := \min(\alpha(S))$ is the smallest element of $S \setminus \{b, m - b + 1\}$ if this set is nonempty; otherwise $b' = m + 1$. Since all distinct elements of S are $\geq b$ and $\neq b$, we have $b' \geq b + 1$.

Step 3: New slope s' and slope set D' . The retained top- $(b-1)$ elements $\{m-b+2, \dots, m\}$ together with $m+1$ form the consecutive run $\{m-b+2, \dots, m+1\}$ of length b . We claim this is the entire new slope, i.e., $s' = b$ and $D' = \{m-b+2, \dots, m+1\}$.

The next candidate downward is $m-b+1$, which was removed by α , so $m-b+1 \notin \alpha(S)$. Hence the top run terminates and $s' = b$, $D' = \{m-b+2, \dots, m+1\}$.

Step 4: Verify $\alpha(S) \in \mathcal{P}_\beta(n)$. We must show $b' > s' = b$ (with $b' \notin D'$ in the boundary case $b' = b+1$), or $b' \geq s' + 2 = b+2$.

From Step 2, $b' \geq b+1$.

Subcase (i): $b' \geq b+2$. Then $b' \geq s' + 2$, second clause of $\mathcal{P}_\beta$ holds.

Subcase (ii): $b' = b+1$. We need $b' \notin D'$. We have $D' = \{m-b+2, \dots, m+1\}$. By Step 0, $m \geq 2b$, so $m-b+2 \geq b+2 > b+1 = b'$. Hence $b' < \min(D')$, so $b' \notin D'$, first clause of $\mathcal{P}_\beta$ holds.

In both subcases, $\alpha(S) \in \mathcal{P}_\beta(n)$. □

Lemma 17. *If $S \in \mathcal{P}_\beta(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n)$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b , slope s , max m , slope set $D = \{m-s+1, \dots, m\}$. Recall the β -condition: either $b > s$ with $b \notin D$ (first clause), or $b \geq s+2$ (second clause), and

$$\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}.$$

Step 0: Key inequality $m \geq 2s+1$. Since $D \subseteq S$ and $b = \min(S)$, we have $b \leq \min(D) = m-s+1$.

- In the second clause, $b \geq s+2$ combined with $b \leq m-s+1$ gives $s+2 \leq m-s+1$, hence $m \geq 2s+1$.
- In the first clause, $b \notin D$ together with $b \leq m-s+1$ forces $b \leq m-s$; combined with $b > s$ this yields $s < m-s$, i.e. $m \geq 2s+1$.

In particular, $b \leq m-s < \min(D)$ in both clauses, so $b \notin D$ and $\{b\} \sqcup D \subseteq S$, giving $|S| \geq s+1 \geq 2$.

Step 1: $\beta(S)$ is a partition of n . The sum change is $-m+s+(m-s) = 0$. For distinctness of the resulting set:

- $s \notin S \setminus \{m\}$: since $b > s$ and $b = \min(S)$, no element of S equals s .
- $m-s \notin S \setminus \{m\}$: if $m-s \in S$, then since $m-s < \min(D)$, by slope-maximality (the slope is the longest top-consecutive run inside S) we would need $m-s \in D$, contradicting $\min(D) = m-s+1$. Also $m-s \neq m$ since $s \geq 1$.
- $s \neq m-s$: by Step 0, $m \geq 2s+1 > 2s$.
- Positivity: $s \geq 1$, and $m-s \geq s+1 \geq 2$.

Step 2: New extremes. The new base is

$$b' := \min(\beta(S)) = \min(b, s) = s,$$

since $b > s$. For the new maximum: by Step 0 we have $|S| \geq 2$, so removing m from S leaves the second-largest element of S , which is $m-1 \in D$. The newly added elements s and $m-s$ are both at most $m-1$ (using $m \geq 2s+1$), so

$$m' := \max(\beta(S)) = m-1.$$

Step 3: New slope and slope set. The shifted top run $\{m-s, m-s+1, \dots, m-1\}$ has length s and lies in $\beta(S)$. We split on the value of $m-2s$.

Case (a): $m \geq 2s+2$. Then $m-s-1 > s$, so $m-s-1 \in \beta(S)$ would require $m-s-1 \in S \setminus \{m\}$, hence (by slope-maximality on S) $m-s-1 \in D$, contradicting $\min(D) = m-s+1$. The top run therefore terminates at $m-s$, giving

$$s' = s, \quad D' = \{m-s, \dots, m-1\}.$$

Case (b): $m = 2s+1$. Then $m-s-1 = s$, which is precisely the new bottom element added by β . The top run extends by one to include it:

$$s' = s+1, \quad D' = \{s, s+1, \dots, 2s\}.$$

Step 4: Verify $\beta(S) \in \mathcal{P}_\alpha(n)$.

Case (a): $m \geq 2s+2$. Here $b' = s$ and $s' = s$, so $b' = s'$ and the second clause $b' \leq s' - 1$ fails. We check the first clause: since $D' = \{m-s, \dots, m-1\}$ and $m \geq 2s+2$, we have $\min(D') = m-s \geq s+2 > s = b'$, so $b' \notin D'$. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$ via the first clause.

Case (b): $m = 2s+1$. Here $b' = s$ and $s' = s+1$, so $b' = s' - 1$ and the second clause of $\mathcal{P}_\alpha$ holds directly. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$.

In both cases, $\beta(S) \in \mathcal{P}_\alpha(n)$. $\square$

Lemma 18. *The composition of maps $\beta \circ \alpha$ is equal to the identity map on $\mathcal{P}_\alpha(n)$. Equivalently, if $S \in \mathcal{P}_\alpha(n)$ then $\beta(\alpha(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b , slope s , max m , slope set D .

Step 1: Compute $\alpha(S)$. By definition,

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}.$$

By Lemma 16, $\alpha(S) \in \mathcal{P}_\beta(n)$. From the proof of that lemma, the data of $\alpha(S)$ is:

- new max $m^\alpha = m+1$,
- new slope $s^\alpha = b$ (the run $\{m-b+2, \dots, m+1\}$),
- new slope set $D^\alpha = \{m-b+2, \dots, m+1\}$.

Step 2: Apply β to $\alpha(S)$. By definition,

$$\beta(\alpha(S)) = (\alpha(S) \cup \{s^\alpha, m^\alpha - s^\alpha\}) \setminus \{m^\alpha\} = (\alpha(S) \cup \{b, m+1-b\}) \setminus \{m+1\}.$$

Step 3: Simplify. Substituting $\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}$:

$$\begin{aligned} \beta(\alpha(S)) &= [((S \setminus \{b, m-b+1\}) \cup \{m+1\}) \cup \{b, m-b+1\}] \setminus \{m+1\} \\ &= [(S \setminus \{b, m-b+1\}) \cup \{b, m-b+1, m+1\}] \setminus \{m+1\} \\ &= (S \setminus \{b, m-b+1\}) \cup \{b, m-b+1\} \\ &= S, \end{aligned}$$

where we used that b and $m-b+1$ are in S (so re-adding them after removal recovers S), and that $m+1 \notin S \setminus \{b, m-b+1\}$ (since $m+1 > m \geq$ every element of S). $\square$

Lemma 19. *The composition of maps $\alpha \circ \beta$ is equal to the identity map on $\mathcal{P}_\beta(n)$. Equivalently, if $S \in \mathcal{P}_\beta(n)$ then $\alpha(\beta(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b , slope s , max m , slope set D .

Step 1: Compute $\beta(S)$.

$$\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}.$$

By Lemma 17, $\beta(S) \in \mathcal{P}_\alpha(n)$. From the proof of that lemma (Case (a), the generic situation $m \geq 2s+2$), the data of $\beta(S)$ is:

- new max $m^\beta = m-1$,
- new base $b^\beta = s$,
- new slope $s^\beta = s$ (the run $\{m-s, \dots, m-1\}$),
- new slope set $D^\beta = \{m-s, \dots, m-1\}$.

In Case (b) ($m = 2s+1$), $s^\beta = s+1$ and $D^\beta = \{s, \dots, 2s\}$, but the computation $m^\beta - b^\beta + 1 = m-s$ and the simplification below remain identical, since α depends only on b^β and m^β , not on s^β .

Step 2: Apply α to $\beta(S)$.

$$\alpha(\beta(S)) = (\beta(S) \setminus \{b^\beta, m^\beta - b^\beta + 1\}) \cup \{m^\beta + 1\} = (\beta(S) \setminus \{s, m-s\}) \cup \{m\}.$$

(Using $b^\beta = s$, $m^\beta = m-1$, so $m^\beta - b^\beta + 1 = m-1-s+1 = m-s$.)

Step 3: Simplify. Substituting $\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}$:

$$\begin{aligned} \alpha(\beta(S)) &= [(S \cup \{s, m-s\}) \setminus \{m\} \setminus \{s, m-s\}] \cup \{m\} \\ &= [(S \setminus \{m\}) \setminus \{s, m-s\}] \cup \{m\}. \end{aligned}$$

Since $s \notin S$ and $m-s \notin S$ (verified in the proof of Lemma 17), removing them from $S \setminus \{m\}$ has no effect:

$$\alpha(\beta(S)) = (S \setminus \{m\}) \cup \{m\} = S,$$

since $m \in S$. □

Lemma 20. *The map $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ is a bijection.*

The map $\beta : \mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$ is a bijection.

Proof. Lemma 16 shows α is well-defined as a map $\mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$, and Lemma 17 shows β is well-defined as a map $\mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$.

Lemma 18 gives $\beta \circ \alpha = \text{id}_{\mathcal{P}_\alpha(n)}$, and Lemma 19 gives $\alpha \circ \beta = \text{id}_{\mathcal{P}_\beta(n)}$.

Hence α and β are mutually inverse bijections. □

Lemma 21.

$$|\mathcal{P}_\alpha(n)| = |\mathcal{P}_\beta(n)|.$$

Proof. A bijection between finite sets implies equal cardinality. Apply this to the bijection $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ from Lemma 20. □

Lemma 22. *The operations α and β change the parity of the number of parts of a partition. More precisely,*

- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$;*
- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$;*

- If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$;
- If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$.

Proof. Membership in $\mathcal{P}_\beta$ for $\alpha(S)$ (and in $\mathcal{P}_\alpha$ for $\beta(S)$) is supplied by Lemmas 16 and 17. It remains to show that the parity of the number of parts is flipped.

Step 1: $|\alpha(S)| = |S| - 1$. By the definition $\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}$:

- we remove two distinct elements (b and $m - b + 1$, distinct as shown in the proof of Lemma 16), reducing the size by 2;
- we add one element ($m + 1$), which is not in $S \setminus \{b, m - b + 1\}$ since $m + 1 > \max(S)$.

Net change: $-2 + 1 = -1$. So $|\alpha(S)| = |S| - 1$, flipping parity.

Step 2: $|\beta(S)| = |S| + 1$. By the definition $\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}$:

- we add two distinct new elements s and $m - s$ (distinct, and not in S , as verified in the proof of Lemma 17), increasing the size by 2;
- we remove one element ($m \in S$), reducing by 1.

Net change: $+2 - 1 = +1$. So $|\beta(S)| = |S| + 1$, flipping parity.

Step 3: Conclusion.

- $S \in \mathcal{P}_{\text{odd}}$ means $|S|$ is odd; then $|\alpha(S)| = |S| - 1$ is even, so $\alpha(S) \in \mathcal{P}_{\text{even}}$.
- $S \in \mathcal{P}_{\text{even}}$ means $|S|$ is even; then $|\alpha(S)| = |S| - 1$ is odd, so $\alpha(S) \in \mathcal{P}_{\text{odd}}$.
- Symmetrically for β with $|\beta(S)| = |S| + 1$.

□

Lemma 23.

$$\begin{aligned} |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)|, \\ |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)|. \end{aligned}$$

Proof. Lemma 22 shows that α restricts to bijections $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n) \rightarrow \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ and $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n) \rightarrow \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ (with inverse β , by Lemma 20). Equal cardinalities follow. □

Lemma 24.

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k, & \text{if } n = \frac{3k^2 + k}{2} \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 1, & \text{if } n = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. **Step 0:** $n = 0$. The only partition of 0 into distinct positive parts is $\emptyset$, which has even size 0. So $p_e(0) = 1$, $p_o(0) = 0$, and $p_e(0) - p_o(0) = 1 = (-1)^0$.

Step 1: Reduction to special partitions. For $n \geq 1$,

$$p_e(n) - p_o(n) = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)|. \quad (1)$$

By Lemma 12 we have

$$|\mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|$$

and

$$|\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\alpha}(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\beta}(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)|.$$

By Lemma 23, $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\alpha}| = |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\beta}|$ and $|\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\alpha}| = |\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\beta}|$. Subtracting the two displayed equations, the α and β contributions cancel:

$$|\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)| - |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)|. \quad (2)$$

Step 2: Compute the special contribution. By Lemma 14, for $n \geq 1$:

- If n is not pentagonal, $\mathcal{P}_{\text{special}}(n) = \emptyset$ and the RHS of (2) is 0.
- If $n = (3k^2 - k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$ with $|S_{-k}| = k$. So the unique special partition contributes +1 to $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\text{special}}| - |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\text{special}}|$ if k is even, and -1 if k is odd. Hence the RHS of (2) equals $(-1)^k$.
- If $n = (3k^2 + k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$ with $|S_k| = k$. Same computation: RHS equals $(-1)^k$.

Combining with (1), the lemma follows. $\square$

3 Jacobi triple product

Theorem 25 (Jacobi triple product). *For all $w \in \mathbb{C}^\times$ and $q \in \mathbb{C}$ with $|q| < 1$:*

$$\prod_{k=1}^{\infty} (1 - q^{2k})(1 - q^{2k-1}w^2)(1 - q^{2k-1}w^{-2}) = \sum_{k=-\infty}^{\infty} (-1)^k w^{2k} q^{k^2}.$$

In q -Pochhammer notation (with $z = -w^2$):

$$(q^2; q^2)_{\infty} (qz; q^2)_{\infty} (q/z; q^2)_{\infty} = \sum_{k \in \mathbb{Z}} (-z)^k q^{k^2}.$$

Many references use $(q; q)_{\infty}(-z; q)_{\infty}(-q/z; q)_{\infty} = \sum_k z^k q^{k(k-1)/2}$, which is equivalent under $q \mapsto q^2$, $z \mapsto -qw^2$.

4 Proof 1: Hermite-style coefficient recurrence

This is the proof in the original outline. The skeleton:

1. Define the partial product $\phi_n(w, q) = \prod_{k=1}^n (1 - q^{2k-1}w^2)(1 - q^{2k-1}w^{-2})$.
2. Derive the functional equation $\phi_n(qw, q) = \phi_n(w, q) \cdot \frac{1 - q^{2n+1}w^2}{q^{2n} - qw^2}$.
3. Expand $\phi_n(w, q) = \sum_{k=-n}^n A_{n,k}(q)w^{2k}$ as a Laurent polynomial.
4. Read off the boundary condition $A_{n,n}(q) = q^{n^2}$ and the recurrence

$$A_{n,k}(q) = A_{n,k-1}(q) q^{2k-1} \frac{1 - q^{2n-2k+2}}{1 - q^{2n+2k}}.$$

5. Solve in closed form:

$$A_{n,k} = \frac{q^{k^2}}{(q^2; q^2)_n} \prod_{s=n-k+1}^n (1 - q^{2s}) \prod_{s=n+k+1}^{2n} (1 - q^{2s}).$$

6. Multiply through by $(q^2; q^2)_n$ and pull out signs to get

$$\prod_{k=1}^n (1 - q^{2k})(1 - q^{2k-1}w^2)(1 - q^{2k-1}w^{-2}) = \sum_{k=-n}^n (-1)^k w^{2k} q^{k^2} B_{n,k}(q),$$

where $B_{n,k}(q) = 1 + O(q^{n-k+1})$.

7. Take $n \rightarrow \infty$ via dominated convergence on $\mathbb{Z}$.

Reference. This is essentially Jacobi's original argument, written up cleanly by many authors; see e.g. Hardy–Wright, *Theory of Numbers*, §19.8, or the lecture notes referenced in the outline.

Cost in Lean. High. The bookkeeping with $A_{n,k}, B_{n,k}$, two interleaved finite products of $1 - q^{2s}$ with s ranges depending on $|k|$, the recurrence with denominators, and the dominated-convergence step on $\mathbb{Z}$ are all individually formalizable but together generate hundreds of small lemmas. Estimate: ~ 1500 – 2000 lines.

5 Proof 2: Euler / q -binomial theorem

The q -binomial theorem. For $|q| < 1$ and $|z| < 1$,

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

Two specializations due to Euler (the “ q -exponentials”):

$$(-z; q)_{\infty} = \sum_{n=0}^{\infty} \frac{q^{n(n-1)/2}}{(q; q)_n} z^n, \quad \frac{1}{(z; q)_{\infty}} = \sum_{n=0}^{\infty} \frac{z^n}{(q; q)_n} \quad (|z| < 1).$$

Andrews' proof (1965). Combine both Euler identities, multiply, and rearrange so the diagonal terms produce $z^k q^{k(k-1)/2}$. A particularly clean modern version is Zhu's *semi-finite* proof (arXiv:2106.16156, 2 pages): for any $m \geq 0$ and $z \neq 0$,

$$\sum_{n=-m}^{\infty} \frac{q^{n(n-1)/2} z^n}{(q^{m+1}; q)_n} = (q; q)_m (-q/z; q)_m (-z; q)_{\infty},$$

proved by reindexing and applying *only* the first Euler identity. Letting $m \rightarrow \infty$ (with z in a compact subset of $\mathbb{C}^{\times}$) gives Jacobi's identity directly.

References.

- G. E. Andrews, *A simple proof of Jacobi's triple product identity*, Proc. Amer. Math. Soc. **16** (1965), 333–334.
- G. E. Andrews, *The Theory of Partitions*, Cambridge, 1976, Ch. 2.
- G. E. Andrews, R. Askey, R. Roy, *Special Functions*, Cambridge, 1999, Ch. 10.
- J.-M. Zhu, *A semi-finite proof of Jacobi's triple product identity*, arXiv:2106.16156.
- A. B. Shukla, *Tiling proofs of Jacobi triple product and Rogers–Ramanujan identities*, arXiv:2006.03878 (gives a combinatorial proof via the q -binomial theorem).

Cost in Lean. High *up front*, low marginal. You must first develop a small theory of q -Pochhammer symbols and prove the q -binomial theorem. After that, the Jacobi triple product is a few-line corollary (modulo handling convergence of bilateral series). The infrastructure pays for itself if you later want Euler's pentagonal number theorem, Rogers–Ramanujan, or any other classical q -identity.

6 Proof 3: Complex analysis via quasi-periodicity

This is the proof in Stein–Shakarchi, *Complex Analysis*, Ch. 10, §1. View both sides as functions of w for $|q| < 1$ fixed; equivalently, view both sides as functions of $z \in \mathbb{C}$ where $w = e^{\pi iz}$ and $q = e^{\pi i\tau}$, $\text{Im } \tau > 0$.

The argument.

1. Define $\Theta(z \mid \tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau + 2\pi i n z}$ (the RHS) and $\Pi(z \mid \tau) = \prod_{m \geq 1} (1 - q^{2m})(1 + q^{2m-1}e^{2\pi iz})(1 + q^{2m-1}e^{-2\pi iz})$ (the LHS, up to $w \mapsto iw$ to match signs).
2. Show both are entire in z (uniform convergence on compact sets; for Θ this is Gaussian decay, for Π this is the $\sum \|1 - a_k\| < \infty$ criterion).
3. Verify both satisfy the *same* two functional equations:

$$F(z+1) = F(z), \quad F(z+\tau) = e^{-\pi i\tau - 2\pi iz} F(z).$$

4. Show both have the *same simple zeros*, located at $z = \frac{1}{2} + \frac{\tau}{2} + \mathbb{Z} + \tau\mathbb{Z}$. For Π this is direct; for Θ it follows from the functional equations plus the fact that the quotient Θ/Π is doubly quasi-periodic and entire, hence constant.
5. Compute the constant by evaluating at one point (e.g., $z = \frac{1}{2}$).

Reference. E. M. Stein, R. Shakarchi, *Complex Analysis*, Princeton, 2003, Chapter 10, §1.1 (“Product formula for the Jacobi theta function”). See also Wikipedia, *Theta function*, and the survey at metaphor.ethz.ch/x/2019/fs/401-4110-19L/sc/talk4.pdf.

Cost in Lean. Moderate. The argument leans on Mathlib's existing complex analysis (locally uniform convergence implies holomorphic limit, Liouville-style uniqueness from doubly periodic entire functions). The hardest formal step is establishing local uniform convergence of the LHS product, plus identifying zero sets. No combinatorial coefficient bookkeeping.

7 Mathlib infrastructure useful for any of the three

Infinite sums and products.

- `Mathlib.Topology.Algebra.InfiniteSum.Defs` – `Summable`, `Multipliable`, `tsum` ($\sum'$), `tprod` ($\prod'$).
- `Mathlib.Topology.Algebra.InfiniteSum.NatInt` – relating $\mathbb{Z}$ -indexed sums to $\mathbb{N}$ -indexed sums (essential for the RHS of the Jacobi identity).
- `Mathlib.Analysis.Normed.Field.InfiniteSum` – `Summable.mul_norm`, Cauchy products in normed rings, dominated convergence for series.
- `Mathlib.Analysis.SpecificLimits.Normed` – `summable_geometric_of_norm_lt_one`, the workhorse for proving $\sum \|q^k\|$ converges.

Infinite products via logarithms (for Proofs 1 and 3).

- `Multipliable.of_summable_log` and friends – a product $\prod a_k$ converges iff $\sum \log a_k$ converges (in a suitable region). Useful when each factor $a_k = 1 - b_k$ with $\sum \|b_k\| < \infty$.
- For each factor of the form $1 - q^{2k-1}w^{\pm 2}$, summability of $\sum \|q^{2k-1}\|$ is geometric.

Complex analysis (for Proof 3).

- `Mathlib.Analysis.Complex.LocallyUniformLimit` – `TendstoLocallyUniformlyOn.differentiableOn`: a locally uniform limit of holomorphic functions is holomorphic. Exactly the tool you need to transport holomorphy from finite partial products to the infinite product.
- `Mathlib.Analysis.Complex.Liouville` – bounded entire is constant; doubly periodic entire is constant.
- `Mathlib.Analysis.Calculus.UniformLimitsDeriv` – derivative of a uniform limit is the limit of derivatives, useful for zero-counting via the argument principle.

What is *not* yet in Mathlib (as of early 2026).

- No q -Pochhammer symbol $(a; q)_n$, $(a; q)_\infty$. (The existing `pochhammer` in Mathlib is the rising factorial, a different object.)
- No q -binomial coefficients or q -binomial theorem.
- No Jacobi theta functions $\vartheta_1, \dots, \vartheta_4$.
- No general theory of doubly periodic / elliptic functions beyond the basics. The Weierstrass $\wp$ -function is partially developed but not the full machinery one would want.

8 Developing q -series in Mathlib: how much work?

If the goal is just Jacobi’s triple product, you do *not* need a full q -series library — you can prove the identity ad hoc by any of the three routes and never mention q -Pochhammer symbols. But if you want a reusable foundation (and the natural framing for Proof 2), here is a realistic estimate.

Minimal viable infrastructure (~ 500 lines).

- Definition: `qPochhammer (a q : R) : ℕ → R` as a finite product, `qPochhammerInf (a q : ℂ) : ℂ` as a `tprod` (with API conditioned on $|q| < 1$).
- Convergence: `Multipliable` for $(a; q)_\infty$ when $|q| < 1$, including continuity in a and q .
- Basic identities: $(a; q)_{n+1} = (a; q)_n(1 - aq^n)$, $(a; q)_\infty = (a; q)_n(aq^n; q)_\infty$, the reflection $(a; q)_\infty / (aq^n; q)_\infty = (a; q)_n$.
- One Euler q -exponential identity (the easier of the two): $(-z; q)_\infty = \sum_n q^{n(n-1)/2} z^n / (q; q)_n$.

This is enough to run Zhu’s two-page semi-finite proof of Jacobi.

Full q -binomial theorem package (~ 1500 – 2000 lines).

- q -binomial coefficients $\binom{n}{k}_q$ and their Pascal-style recursions.
- The finite q -binomial theorem $(a; q)_n = \sum_k \binom{n}{k}_q (-a)^k q^{\binom{k}{2}}$.
- The Cauchy identity / infinite q -binomial theorem $\sum_n \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_\infty}{(z; q)_\infty}$ for $|z| < 1$.
- Both Euler identities as limits ($a \rightarrow 0$, $a \rightarrow \infty$).
- Heine’s transformation (a useful but optional generalization).

This gets you Jacobi’s triple product, Euler’s pentagonal number theorem, and roughly half of Andrews’ *Theory of Partitions* Ch. 2 as one-line corollaries.

Beyond ($\sim 3000+$ lines). Rogers–Ramanujan, the Rogers–Fine identity, basic hypergeometric series ${}_r\phi_s$, the theory of partitions with restrictions, etc. This is an entire subfield; nobody has formalized it.

Recommended order of attack.

1. Define $(a; q)_n$ as a polynomial, develop algebraic identities (q -Pascal, finite q -binomial).
2. Define $(a; q)_\infty$ as a `tprod`, prove `Multipliable` for $|q| < 1$, prove continuity.
3. Prove $(-z; q)_\infty = \sum q^{\binom{n}{2}} z^n / (q; q)_n$ by manipulating partial sums and passing to the limit.
4. Apply Zhu’s argument to get Jacobi’s triple product.
5. (Optional) Prove the second Euler identity and the full q -binomial theorem.

9 Recommendation

If the *single* goal is Jacobi’s triple product in Lean, my ranking is:

1. **Proof 3 (complex analysis)**, if you are comfortable with Mathlib’s complex analysis layer. The functional-equation argument is short, leverages existing Mathlib infrastructure, and produces clean intermediate lemmas (holomorphy of theta, periodicity, zero set) that are independently useful. Estimate: ~ 1000 – 1500 lines.

2. **Proof 2 (Zhu’s semi-finite proof)**, if you are willing to develop the minimal q -Pochhammer infrastructure (~ 500 extra lines) and want a reusable foundation. The main proof is then very short. Total: ~ 1200 lines.
3. **Proof 1 (Hermite-style)**, if you want a self-contained elementary proof that uses no analysis beyond geometric series. The longest of the three but conceptually the most direct. Total: ~ 1800 lines.

The right choice depends on what else you plan to formalize. For Jacobi alone: Proof 3. For a q -series library: Proof 2. For pedagogical clarity and minimal prerequisites: Proof 1.

10 Euler’s pentagonal number theorem from Jacobi

Theorem 26 (Euler). *For $|q| < 1$,*

$$\prod_{n=1}^{\infty} (1 - q^n) = \sum_{k=-\infty}^{\infty} (-1)^k q^{k(3k-1)/2} = 1 + \sum_{k=1}^{\infty} (-1)^k (q^{k(3k-1)/2} + q^{k(3k+1)/2}).$$

Proof. Use Jacobi’s triple product in the symmetric two-variable form

$$\prod_{n=1}^{\infty} (1 - x^n)(1 - x^{n-1}z)(1 - x^n z^{-1}) = \sum_{k \in \mathbb{Z}} (-1)^k z^k x^{k(k-1)/2}, \quad (\star)$$

valid for $|x| < 1$ and $z \in \mathbb{C}^\times$. (This is equivalent to Theorem 25 via $x = q^2$, $z = qw^2$; we use this form because it avoids the w^{-2} asymmetry of the original statement.)

Substitute $x = q^3$ and $z = q$. The three factors on the left become

$$\prod_{n=1}^{\infty} (1 - q^{3n})(1 - q^{3n-3} \cdot q)(1 - q^{3n} \cdot q^{-1}) = \prod_{n=1}^{\infty} (1 - q^{3n})(1 - q^{3n-2})(1 - q^{3n-1}).$$

The three index sets $\{3n : n \geq 1\}$, $\{3n - 2 : n \geq 1\}$, $\{3n - 1 : n \geq 1\}$ partition $\mathbb{Z}_{\geq 1}$ (they are the residue classes $0, 1, 2 \pmod{3}$), so the product collapses to

$$\prod_{m=1}^{\infty} (1 - q^m).$$

The right side becomes

$$\sum_{k \in \mathbb{Z}} (-1)^k q^k \cdot q^{3k(k-1)/2} = \sum_{k \in \mathbb{Z}} (-1)^k q^{k+3k(k-1)/2} = \sum_{k \in \mathbb{Z}} (-1)^k q^{k(3k-1)/2},$$

since $k + 3k(k-1)/2 = k(3k-1)/2$. Splitting the sum into $k \geq 1$, $k = 0$, $k \leq -1$ and using $(-k)(3(-k)-1)/2 = k(3k+1)/2$ gives the second form. $\square$

Formalization in Lean. Given Jacobi’s triple product in the two-variable form $(\star)$, this is a short proof:

1. **Substitution** (~ 10 lines): apply Jacobi with $x := q^3$, $z := q$. The hypothesis $|x| < 1$ follows from $|q| < 1$ by $\|q^3\| = \|q\|^3 < 1$.

2. **Index partitioning** (~ 30 lines): prove the bijection $\mathbb{N}_{>0} \times \{0, 1, 2\} \rightarrow \mathbb{N}_{>0}$ sending $(n, r) \mapsto 3n - r$ (suitably offset), then use `Multipliable.tprod_eq_of_hasProd` or rewrite the three products as one via a `tprod` change of variables. Mathlib's `tprod_sigma`-style lemmas handle this.
3. **Exponent arithmetic** (~ 20 lines): show $k + 3k(k-1)/2 = k(3k-1)/2$ for $k \in \mathbb{Z}$. Since $k(3k-1)$ is always even (one of $k, 3k-1$ is), use `Int.ediv` and prove the identity via `omega` or `ring` after multiplying through by 2.

Total: ~ 60 lines once Jacobi is available.

11 The q -binomial theorem: statement and proof

11.1 Statement

Define the *finite* q -Pochhammer symbol

$$(a; q)_n = \prod_{k=0}^{n-1} (1 - aq^k), \quad (a; q)_0 = 1,$$

the q -factorial $[n]_q! = (q; q)_n / (1 - q)^n$, and the *Gaussian binomial coefficient*

$$\binom{n}{k}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} \quad (0 \leq k \leq n).$$

By convention $\binom{n}{k}_q = 0$ for $k < 0$ or $k > n$. These are polynomials in q with integer coefficients (this is part of what needs to be proved).

Theorem 27 (Finite q -binomial theorem). *In $\mathbb{Z}[q, z]$, for every $n \geq 0$,*

$$\prod_{k=0}^{n-1} (1 + zq^k) = \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k.$$

Theorem 28 (Infinite q -binomial / Cauchy identity). *For $|q| < 1$ and any $a, z \in \mathbb{C}$ with $|z| < 1$,*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

The two Euler identities follow from Theorem 28 by specializing $a = 0$ (gives $1/(z; q)_{\infty} = \sum z^n / (q; q)_n$) and by sending $a \rightarrow \infty$ along $a = -y/q^N$ with $N \rightarrow \infty$, then renaming (gives $(-z; q)_{\infty} = \sum q^{\binom{n}{2}} z^n / (q; q)_n$).

11.2 Proof of the finite q -binomial theorem

This proof is purely algebraic (no analysis) and is the version best suited for Lean. It uses only induction on n and the q -Pascal identity.

Lemma 29 (q -Pascal). *For all $n, k \in \mathbb{N}$,*

$$\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q.$$

(Equivalently, with $k \geq 1$: $\binom{n+1}{k}_q = \binom{n}{k}_q + q^{n-k+1} \binom{n}{k-1}_q$. A symmetric form $\binom{n+1}{k+1}_q = q^{k+1} \binom{n}{k+1}_q + \binom{n}{k}_q$ also holds; we use the first.)

Proof. This is the recursive defining equation of $\binom{n}{k}_q$ in Definition 33, hence `rfl` in Lean. Equivalently, by direct computation from the quotient form $\binom{n}{k}_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$ (valid when $k \leq n$ and the denominators are nonzero): $\binom{n}{k}_q + q^{n-k+1} \binom{n}{k-1}_q = \binom{n}{k-1}_q \cdot \frac{(1-q^{n-k+1}) + q^{n-k+1}(1-q^k)}{1-q^k} = \binom{n}{k-1}_q \cdot \frac{1-q^{n+1}}{1-q^k} = \binom{n+1}{k}_q$, recovering the off-by-one form. $\square$

Proof of Theorem 27. Induction on n , following the Lean proof `qSeries.qBinom_finite_thm` in `Series/qSeries.lean`. The case $n = 0$ gives $1 = 1$.

For the inductive step, set $P_n(z) := \prod_{k=0}^{n-1} (1 + zq^k)$ and $Q_n(z) := \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k$, so the theorem reads $P_n = Q_n$. By `Finset.prod_range_succ` and the inductive hypothesis $P_n = Q_n$,

$$P_{n+1}(z) = P_n(z) (1 + zq^n) = Q_n(z) (1 + zq^n),$$

so it suffices to show $Q_{n+1}(z) = Q_n(z) (1 + zq^n)$. We decompose $Q_{n+1}(z) = \sum_{k=0}^{n+1} q^{\binom{k}{2}} \binom{n+1}{k}_q z^k$ in four explicit steps (mirroring `stepB-stepD` of the Lean proof).

Step A (peel off $k = 0$ via `Finset.sum_range_succ'`). Using $\binom{n+1}{0}_q = 1$,

$$Q_{n+1}(z) = 1 + \sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n+1}{k+1}_q z^{k+1}.$$

Step B (q -Pascal split). Apply Lemma 29 inside the sum and split via `Finset.sum_add_distrib`:

$$\sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n+1}{k+1}_q z^{k+1} = \underbrace{\sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n}{k+1}_q z^{k+1}}_{S_1} + \underbrace{\sum_{k=0}^n q^{\binom{k+1}{2} + n - k} \binom{n}{k}_q z^{k+1}}_{S_2}.$$

Step C ($S_2 = zq^n Q_n(z)$). The identity $\binom{k+1}{2} = \binom{k}{2} + k$ (helper `choose_two_succ`) and $k + (n - k) = n$ (valid because $k \leq n$, by `Nat.sub_add_cancel`) give $q^{\binom{k+1}{2} + n - k} z^{k+1} = zq^n \cdot q^{\binom{k}{2}} z^k$, hence

$$S_2 = zq^n \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k = zq^n Q_n(z).$$

Step D ($1 + S_1 = Q_n(z)$). Reindex $j = k + 1$ to identify $S_1 = \sum_{j=1}^{n+1} q^{\binom{j}{2}} \binom{n}{j}_q z^j$, then adjoin the $j = 0$ term 1 (using `Finset.sum_range_succ'` in reverse):

$$1 + S_1 = \sum_{j=0}^{n+1} q^{\binom{j}{2}} \binom{n}{j}_q z^j.$$

The $j = n + 1$ term vanishes by Lemma 34 ($\binom{n}{n+1}_q = 0$), reducing this to $Q_n(z)$.

Combining the four steps, $Q_{n+1}(z) = (1 + S_1) + S_2 = Q_n(z) + zq^n Q_n(z) = Q_n(z) (1 + zq^n)$. The final algebraic identity is closed in Lean by `linear_combination`. $\square$

11.3 Passing to the infinite case

Proof sketch of Theorem 28. The finite form of Theorem 28 is the analogous identity

$$\sum_{k=0}^n \frac{(a; q)_k}{(q; q)_k} z^k \cdot \frac{(q^{n-k+1}; q)_k}{(q^{n-k+1}; q)_k} = \frac{(az; q)_n}{(z; q)_n} \cdot (\text{correction})$$

— one needs to be careful here; the cleanest finite version is due to Cauchy and reads, for $|z| < 1$,

$$\sum_{k=0}^n \binom{n}{k}_q \frac{(a; q)_k}{(q; q)_k} z^k \xrightarrow{n \rightarrow \infty} \sum_{k=0}^{\infty} \frac{(a; q)_k}{(q; q)_k} z^k.$$

The classical route is:

1. Prove the finite identity $(az; q)_n = \sum_{k=0}^n \binom{n}{k}_q (-a)^k q^{\binom{k}{2}} z^k$ · (telescope) — a polynomial identity provable by the same induction as Theorem 27.
2. Pass to the limit $n \rightarrow \infty$ using $(q; q)_k / (q; q)_n \rightarrow 1 / (q^{k+1}; q)_{\infty}$ and dominated convergence (each $|q^{\binom{k}{2}} z^k| \leq |z|^k$ is summable when $|z| < 1$).

□

Avoiding the infinite version entirely. For Jacobi’s triple product via Zhu’s semi-finite proof, you do not actually need Theorem 28. You only need the *first Euler identity*

$$(-z; q)_{\infty} = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}}}{(q; q)_n} z^n,$$

which is the $n \rightarrow \infty$ limit of Theorem 27 after rewriting $\prod_{k=0}^{n-1} (1 + zq^k)$ as $(-z; q)_n$ and dividing by suitable factors. This limit is gentler than the full Cauchy identity.

11.4 Formalization plan in Lean

Definitions.

- `qPochhammer (a q : R) : ℕ → R`, defined recursively:
`qPochhammer a q 0 = 1,`
`qPochhammer a q (n+1) = qPochhammer a q n * (1 - a * q^n).`
 Use `Polynomial.eval` for the abstract version in $R[X]$, or work directly in $\mathbb{C}$.
- `qBinom (n k : ℕ) (q : R) : R` as `qPochhammer q q n / (qPochhammer q q k * qPochhammer q q (n - k))` when $k \leq n$, else 0. To stay in $\mathbb{Z}[q]$, define instead as a polynomial via the q -Pascal recurrence and prove the closed form is well-defined.

Recommended approach: define q -binomial coefficients via the recurrence, not the quotient. The quotient definition introduces division-by-zero conditions (the denominator $(q; q)_k$ vanishes at roots of unity). The recurrence-based definition avoids this entirely:

$$\begin{aligned} \text{qBinom } 0 \ 0 \ q &= 1; \text{qBinom } 0 \ (k+1) \ q = 0; \text{qBinom } (n+1) \ 0 \ q = 1; \\ \text{qBinom } (n+1) \ (k+1) \ q &= \text{qBinom } n \ (k+1) \ q + q^{n-k} * \text{qBinom } n \ k \ q. \end{aligned}$$

Then `qBinom n k` is a polynomial in q with integer coefficients by construction. The closed-form equality with $(q; q)_n / ((q; q)_k (q; q)_{n-k})$ becomes a *theorem* provable for q outside the relevant roots of unity (or as an identity of rational functions).

Step-by-step Lean roadmap.

1. Define `qPochhammer` (Mathlib's `Finset.prod_range` suffices: `qPochhammer a q n := ∏ k ∈ range n, (1 - a * q^k)`). ~ 30 lines.
2. Prove basic identities: `qPochhammer a q (n+1) = qPochhammer a q n * (1 - a * q^n)` (this is `Finset.prod_range_succ`); the recursion the other way; multiplicativity. ~ 50 lines.
3. Define `qBinom` by recursion as above; prove symmetry $\binom{n}{k}_q = \binom{n}{n-k}_q$ by induction. ~ 100 lines.
4. Prove the closed-form formula $\binom{n}{k}_q \cdot (q; q)_k \cdot (q; q)_{n-k} = (q; q)_n$ as an identity of polynomials. This is the bridge between the recursive definition and the quotient definition. ~ 100 lines.
5. Prove the finite q -binomial theorem (Theorem 27) by induction, following the proof above. ~ 80 lines.
6. Define `qPochhammerInf` as a `tprod`; prove `Multipliable` for $|q| < 1$ using $\sum \|q^k\| < \infty$. ~ 100 lines.
7. Prove the first Euler identity $(-z; q)_\infty = \sum q^{\binom{n}{2}} z^n / (q; q)_n$ for $|q| < 1$ and z in any bounded set, by taking $n \rightarrow \infty$ in Theorem 27. The convergence step uses dominated convergence with the bound $|q^{\binom{n}{2}}| \leq |q|^{n(n-1)/2}$. ~ 200 lines.
8. (Optional) Prove the full Cauchy identity (Theorem 28). ~ 150 extra lines.

Total for steps 1–7 (sufficient for Zhu's proof of Jacobi): ~ 650 lines. Adding Zhu's argument (~ 100 lines) plus the pentagonal corollary (~ 60 lines) brings the total Jacobi-via- q -binomial route to ~ 800 lines plus whatever scaffolding the bilateral series on the RHS requires.

11.5 Existing Mathlib infrastructure

Algebraic foundations.

- `Mathlib.Algebra.BigOperators.Basic`, `Mathlib.Algebra.BigOperators.Ring` – the $\prod_{k \in s, f k}$ and $\sum_{k \in s, f k}$ machinery, including `Finset.prod_range_succ`, `Finset.prod_range_succ'`, `Finset.prod_Ico_consecutive`, `Finset.prod_bij`.
- `Mathlib.Algebra.Polynomial.Basic` – if you want to formalize q -binomials as elements of $\mathbb{Z}[q]$ rather than as functions of q , this gives `Polynomial`, `Polynomial.coeff`, `Polynomial.eval`.
- `Mathlib.Algebra.Polynomial.Induction` – `Polynomial.induction_on'` for proving polynomial identities by reducing to monomials.
- `Mathlib.Combinatorics.Choose.Basic` – ordinary binomial coefficients `Nat.choose`, useful as a guide and for the $q \rightarrow 1$ limit (not needed for the proof itself).

Analytic foundations (for the infinite case).

- `Mathlib.Topology.Algebra.InfiniteSum.Defs` – `Summable`, `Multipliable`, `tsum` ($\sum'$), `tprod` ($\prod'$).
- `Mathlib.Analysis.Normed.Field.InfiniteSum` – `Summable.mul_norm`, Cauchy products, dominated convergence for sums.
- `Mathlib.Analysis.SpecificLimits.Normed` – `summable_geometric_of_norm_lt_one`, the workhorse summability lemma.
- `Mathlib.Analysis.Normed.Group.InfiniteSum` – `tsum_geometric_of_norm_lt_one` and friends.

What is missing (as of early 2026).

- No `qPochhammer` or `qPochhammerInf`.
- No `qBinom` or Gaussian binomial coefficient. (Mathlib’s existing `pochhammer` is the *rising factorial* $x(x+1)\cdots(x+n-1)$, a different and unrelated object despite the name.)
- No q -Pascal recurrence, no finite or infinite q -binomial theorem, no Euler identities.

This means the entire q -series scaffold needs to be developed from scratch. The good news is that there are no foundational gaps: every step rests on Mathlib infrastructure that already exists. The work is purely additive.

11.6 Why this is worth it

Once the ~ 650 lines of q -series infrastructure are in place, the following all become short corollaries:

- Jacobi’s triple product (Zhu’s proof: ~ 100 lines).
- Euler’s pentagonal number theorem (specialization: ~ 60 lines).
- Gauss’s identities for $\sum q^{n(n+1)/2}$ and $\sum (-1)^n q^{n^2}$ (specializations: ~ 50 lines each).
- The generating function for partitions $\sum p(n)q^n = 1/(q; q)_\infty$ (immediate).
- Generating functions for partitions into distinct parts, odd parts, etc. (Euler’s theorem: ~ 100 lines).

By contrast, formalizing any one of these via a bespoke proof would cost ~ 1000 – 2000 lines and produce no reusable infrastructure. The q -series route amortizes the cost.

12 The q -Pochhammer symbol and Gaussian binomial coefficient (formalized)

This section records the foundational definitions and lemmas that have been formalized in Lean for the q -series approach to Jacobi’s triple product. The corresponding source file is `Series/qSeries.lean` in the `Series` Lean library, under the namespace `qSeries`. Throughout this section R is a commutative ring and $a, q \in R$.

12.1 The q -Pochhammer symbol

Definition 30 (Finite q -Pochhammer symbol). For $n \in \mathbb{N}$, the finite q -Pochhammer symbol is

$$(a; q)_n := \prod_{k=0}^{n-1} (1 - a q^k).$$

In Lean, this is implemented as $\prod k \in \text{Finset.range } n, (1 - a * q^k)$, specialising to $(a; q)_0 = 1$ (the empty product).

Lemma 31 (Base case). $(a; q)_0 = 1$.

Proof. The empty product is 1 (`Finset.prod_range_zero`). □

Lemma 32 (Recurrence). For all $n \in \mathbb{N}$,

$$(a; q)_{n+1} = (a; q)_n \cdot (1 - a q^n).$$

Proof. Immediate from `Finset.prod_range_succ`. □

12.2 The Gaussian binomial coefficient

Definition 33 (Gaussian binomial coefficient). The Gaussian binomial coefficient $\binom{n}{k}_q \in R$ is defined by recursion on n and k :

$$\begin{aligned} \binom{0}{0}_q &= 1, \quad \binom{0}{k+1}_q = 0, \\ \binom{n+1}{0}_q &= 1, \quad \binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q. \end{aligned}$$

This is the q -Pascal recurrence in the form stated in Lemma 29 (with the index shift $k \mapsto k+1$). By construction $\binom{n}{k}_q$ is a polynomial in q with integer coefficients, defined for every pair $(n, k) \in \mathbb{N}^2$ (no division, no roots-of-unity exclusion).

Lemma 34 (Vanishing above the diagonal). If $n < k$, then $\binom{n}{k}_q = 0$.

Proof. Induction on n and k following the four-case recursive definition. The non-trivial case is $(n+1, k+1)$ with $n < k$: the recurrence gives $\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q$, and both inductive hypotheses (at $(n, k+1)$ and (n, k) , available because $n < k+1$ and $n < k$ respectively) yield 0. □

Lemma 35 (Diagonal). $\binom{n}{n}_q = 1$ for every $n \in \mathbb{N}$.

Proof. Induction on n . The case $n = 0$ is by definition. For the step,

$$\binom{n+1}{n+1}_q = \binom{n}{n+1}_q + q^{n-n} \binom{n}{n}_q = 0 + 1 \cdot 1 = 1,$$

using Lemma 34 for the first term and the inductive hypothesis for the second. □

Lemma 36 (Closed form for $\binom{n}{k}_q$). For all $k \leq n$,

$$\binom{n}{k}_q \cdot (q; q)_k \cdot (q; q)_{n-k} = (q; q)_n.$$

Proof. Induction on n .

Base case $n = 0$. Forces $k = 0$ and the identity is $1 \cdot 1 \cdot 1 = 1$.

Boundary $k = 0$. Reduces to $1 \cdot 1 \cdot (q; q)_n = (q; q)_n$.

Boundary $k = n + 1$. By Lemma 35 we have $\binom{n+1}{n+1}_q = 1$, and $(q; q)_0 = 1$, so both sides equal $(q; q)_{n+1}$.

Inductive step $k + 1 \leq n$. Apply the q -Pascal recurrence

$$\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q$$

and the factorisations

$$(q; q)_{n-k} = (q; q)_{n-(k+1)} \cdot (1 - q q^{n-(k+1)}), \quad (q; q)_{k+1} = (q; q)_k \cdot (1 - q q^k),$$

together with the power identities $q \cdot q^{n-(k+1)} = q^{n-k}$ and $q^{n-k} \cdot (q \cdot q^k) = q \cdot q^n$, which use $k \leq n$ via `Nat.sub_add_cancel`. The two inductive hypotheses at $(n, k+1)$ and (n, k) then combine into the required identity, dispatched in Lean by `linear_combination`. $\square$

12.3 The finite q -binomial theorem (step 5)

Theorem 27 is formalised as `qSeries.qBinom_finite_thm` in `Series/qSeries.lean`. The Lean proof follows exactly the four-step strategy spelled out in the proof attached to Theorem 27 above: peel off $k = 0$, apply the q -Pascal recurrence (Lemma 29) inside the sum, simplify the second piece to $z q^n Q_n$, and recombine the first piece via $\binom{n}{n+1}_q = 0$ (Lemma 34).

Remark. Together with Lemma 32, Lemma 34, and Lemma 36, Theorem 27 realises steps 1, 2, 3, 4, and 5 of the formalisation roadmap in Section 11 (“Formalization plan in Lean”). Step 3 is partially completed: the recursive definition is in place (Definition 33) and the closed-form bridge to the quotient description is established (Lemma 36); the symmetry identity $\binom{n}{k}_q = \binom{n}{n-k}_q$ remains as future work. Steps 6–8 (infinite q -Pochhammer products, the first Euler identity, and the full Cauchy identity) are not yet formalised.